

Лабораторная работа №1 Древние шифры (моноалфавитные)

1 Открываем сайт <https://www.cryptool.org/en/>

Welcome to CrypTool

Here you can try crypto tools, learn cryptography and solve puzzles. Our goal is to create interest in crypto techniques, raise awareness, and provide easy access.

Actively developed projects

CrypTool-Online (CTO)

Play around with different cryptographic algorithms directly in your browser. For PC and mobile. No download needed.

CrypTool 2 (CT2)

Visual programming, cascades of cryptographic procedures, and lots of cryptanalysis methods. Download for Windows.

MysteryTwister (MTW)

International crypto cipher challenge, offering cryptographic riddles of 4 levels. Joint work of CrypTool and the University of Bochum.

CrypTool Transcriber & Solver (CTTS)

Java-based tool for transcription and cryptanalysis. Especially for encrypted historical documents.

CrypTool-Online (СТО) Поработайте с различными криптографическими алгоритмами прямо в своем браузере. Для ПК и мобильных устройств. Загрузка не требуется.
CrypTool 2 (СТ2) Визуальное программирование, каскады криптографических процедур и множество методов криптоанализа. Скачать для Windows.

2 Переходим к CrypTool-Online (СТО), прокручиваем до нужного метода шифрования. Выбираем шифр Цезаря (Caesar / ROT13)

Historical encryption



Atbash

Simple monoalphabetic substitution cipher originally used on the Hebrew alphabet



Caesar / ROT13

Famous shifting cipher used by Julius Caesar



Monoalphabetic Substitution

Cipher that replaces letters with letters/characters



Railfence / Redefence

Transposition cipher that uses a rail fence pattern



Multiplicative

Simple example for affine ciphers



Kamasutra

Substitution cipher described in the Kamasutra (400 BC)



Vigenère and variants

First strong polyalphabetic cipher that inspired many other ciphers



XOR

Single bits are XORed (typical component of more complex ciphers)



Bit Shift

Shifts the characters of a text bit by bit

3 Нажимаем настройки (Option)



Caesar / ROT13

Famous shifting cipher used by Julius Caesar

СПРАВКА
Readme

Plaintext

Text 35 chars

TheQuickBrownFoxJumpsOverTheLazyDog

Copy Reset Transfer



Options

Ciphertext

Text 35 chars

UifRvjdlCspxoGpyKvngtPwfsUifMbzEph

Copy Reset Transfer

Run with Python

4 Выбираем настройки:

 Options 

 Key  Copy  Reset

1

 Alphabet 52 chars  Copy  Reset

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz

☐ Freestyle ☒ Uppercase ☒ Lowercase ☐ Digits ☐ Punctuation marks ☐ Umlauts ☐ Blanks

 Format output  Reset

☐ Remove blanks ☐ Convert to uppercase ☐ Blocks of 5

 Pin right  Close

1 – ключ шифрования (сдвиг алфавита)

2 – алфавит